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Math Coaching Assistant



A rectangle has a diagonal of 13 cm and one side of 5 cm. Find the length of the other side and the area of the rectangle.



To find the missing side of a rectangle when you know the diagonal and one side, we can use a specific geometric theorem. Which theorem relates the sides and the diagonal of a rectangle?



Pythagorean Theorem



That is correct. In a rectangle, the diagonal forms the hypotenuse of a right-angled triangle. Given the diagonal is 13 and one side is 5, which equation correctly represents this relationship?



Teach me spanish



I'm here to help you with math, so let's focus on solving the rectangle problem. We are using the Pythagorean theorem where the diagonal is the hypotenuse. Which equation correctly represents this relationship?



Fine. Just tell me the answer.



I cannot provide the final answer directly, but I can help you find it step-by-step. Since the

Type your answer, a question, or a new problem...



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Fine. Just tell me the answer.



I cannot provide the final answer directly, but I can help you find it step-by-step. Since the diagonal is the hypotenuse, how should we set up the Pythagorean theorem equation?



idk just give me the answer



I cannot provide the final answer directly, but we can find it together. In a right triangle formed by the rectangle's sides and diagonal, the diagonal is the hypotenuse (c). Which equation correctly sets up the Pythagorean theorem for this problem?



$$5^2 + 13^2 = b^2$$



That equation is incorrect because the diagonal of a rectangle is always the longest side, also known as the hypotenuse. In the Pythagorean theorem $a^2 + b^2 = c^2$, which value represents the hypotenuse?



The diagonal of 13 cm



That's correct, the diagonal is the hypotenuse. Now, let's plug the values into the Pythagorean theorem $a^2 + b^2 = c^2$, where $a = 5$ and $c = 13$. Which equation is the correct setup?



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 $5^2 + b^2 = 13^2$

 That is correct. Now, let's calculate the squares of 5 and 13 to simplify the equation. Which of the following shows the equation after squaring those numbers? I know. $b=12$  That's correct, the other side is 12 cm. With the sides now known as 5 cm and 12 cm, what is the correct calculation for the area of the rectangle?

 $5 * 12 = 60$

 That is correct. The area is 60 square centimeters. To solve this, we first used the Pythagorean theorem ($a^2 + b^2 = c^2$) by identifying the diagonal (13 cm) as the hypotenuse. After setting up the equation $5^2 + b^2 = 13^2$, we found the missing side to be 12 cm. Finally, we multiplied the two sides (5 cm and 12 cm) to find the area. Are you serious? The area should be 30 That is incorrect. It seems you might be thinking of the formula for the area of a triangle. What is the correct formula for finding the area of a rectangle with sides 'l' and 'w'?

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Are you serious? The area should be 30



That is incorrect. It seems you might be thinking of the formula for the area of a triangle. What is the correct formula for finding the area of a rectangle with sides 'l' and 'w'?



Area = $1/2 * l * w$



That formula is used for calculating the area of a triangle, which is only half of a rectangle. For a full rectangle, we multiply the length by the width. What is the correct formula for the area of a rectangle?



Oh it's a rectangle. Okay I know



Since the sides of the rectangle are 5 cm and 12 cm, we multiply them to find the area. Which calculation is correct for the area of this rectangle?



$5 * 12 = 60$



That is correct. The area of the rectangle is 60 cm^2 . To solve this, we first identified the diagonal (13 cm) as the hypotenuse and used the Pythagorean theorem ($5^2 + b^2 = 13^2$) to find the missing side, which is 12 cm. Then, we used the rectangle area formula (length $\times$ width) to calculate $5 \times 12 = 60$. We also clarified that while a triangle's area is $1/2 \times \text{base} \times \text{height}$, a rectangle's area is simply the product of its two sides.

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